

Performance of a multi-sensor wearable system for validating gait assessment: preliminary results on patients and healthy

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Introduction: Real-world (RW) gait analysis is becoming increasingly important since in these conditions an individual's mobility performance can be measured, in addition to what happens in a standardised lab environment, where mobility capacity is assessed. This is even more relevant in the presence of a motor disability, when continuous daily-life monitoring can unveil limitations in movement, better stratify disease progression and the efficacy of medical treatments [1]. Wearable inertial measurement units (IMUs) represent the most valid solution to tackle this challenge, but their validation requires RW gait data from a reliable Gold Standard (GS). This study evaluates the performance of a wearable multi-sensor system (INDIP) which was specifically conceived to be used as a RW GS [2]. In particular, the INDIP performance in estimating digital mobility outcomes (DMOs) was assessed against a stereophotogrammetric (SP) system, on healthy older adults and patients from five different cohorts, following a purposely designed experimental protocol [3].

Methods: The INDIP system includes three IMUs (sampling frequency (fs) = 100 Hz) positioned on lower back and feet, two pressure insoles (16 sensing elements, fs = 100 Hz) and two time-of-flight distance sensors (fs = 50 Hz, range = 0.2 m) [2]. Experiments were carried out on 107 participants from six different cohorts (Healthy Adults (HA), Parkinson's Disease (PD), Multiple Sclerosis (MS), Chronic Obstructive Pulmonary Disease (COPD), Congestive Heart Failure (CHF), Proximal Femoral Fracture (PFF)) while performing a complex eleven-task protocol [3]. Walking bouts (WBs) were obtained from the INDIP data, according to the flowchart described in [2]. Then, for each WB, relevant DMOs were computed.

Table 1

Results for walking speed. STD: Standard Deviation.

Cohort	Participants number	WBs number	Walking Speed		
			Mean ± STD (INDIP)	Mean ± STD (SP)	Median Relative Error (%)
HA	20	237	0.85 ± 0.30 m/s	0.85 ± 0.30 m/s	2.28 %
PD	20	241	0.74 ± 0.30 m/s	0.72 ± 0.30 m/s	2.42 %
MS	19	238	0.79 ± 0.30 m/s	0.79 ± 0.30 m/s	2.55 %
COPD	17	220	0.81 ± 0.30 m/s	0.81 ± 0.31 m/s	2.02 %
CHF	10	105	0.85 ± 0.33 m/s	0.80 ± 0.29 m/s	2.58 %
PFF	17	193	0.64 ± 0.33 m/s	0.64 ± 0.32 m/s	2.32 %

Accuracy of the estimated DMOs was evaluated at cohort level in terms of median relative error (%). For sake of brevity, here we only report walking speed results.

Results: A total of 103 subjects were considered for the analysis (1234 WBs in total). Results at cohort level for WB walking speed are presented in Table 1. Median relative errors ranged from 2.02% to 2.58%.

Discussion: Results reported for the WB walking speed showed very low median relative errors across all cohorts. For each cohort, the distributions of values from the INDIP and SP were very similar as shown by the means and standard deviations in Table 1. Such low differences are excellent, especially when considering the complexity of the tasks included in the experimental protocol. In conclusion, the INDIP system is an accurate reference system for RW gait analysis.

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An innovative integrated platform of telerehabilitation for patients with chronic diseases: development and initial implementation

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Introduction: Although physical exercise is recommended to maintain health and to counteract the deleterious effects of chronic diseases, home exercising is rarely performed autonomously by patients. Hence, there is urgency to introduce and validate rehabilitation protocols based on a combination of technological solutions and specific health requirements in order to guarantee continuity and quality of care to patients with chronic diseases [1]. The aim of this presentation is to describe the development and initial implementation of an integrated platform of telerehabilitation, based on information and communication technologies. The platform is the product of a Lombardy Region grant (Regione Lombardia, Programma Operativo Regionale 2014-2020, "Bando Call HUB Ricerca e Innovazione") to Istituti Clinici Scientifici Maugeri IRCCS.

Methods: Preliminary results have been obtained in five healthy subjects testing the feasibility of the integrated platform in a lab setting (Fig. 1). The platform includes an @home fully-integrated kit composed of: 1) wearable inertial devices, 2) digitally-controlled resistance band, 3) time-of-flight range camera 4) mini-PC with exergames. The @home kit is connected with: 4) cloud-based software systems for programming and monitoring exercises, 5) intelligent conversational agents (chatbot) and virtual coach on personal smartphones. The exergames are based on Otago- inspired exercises [2], including strength, balance, aerobic and cognition. The hardware of the exergames consist of a mini-PC, to

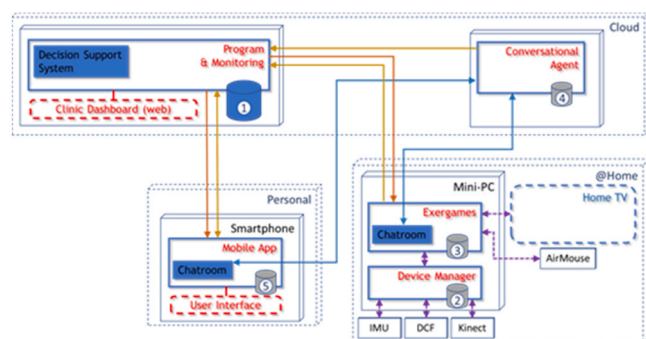


Fig. 1. Architectural schema of the integrated platform showing the interconnections between the patient (@home kit), the cloud-based systems and the intelligent conversational agents. IMU, inertial measurement unit; DCF, digitally-controlled resistance band.

be connected to a home TV set, eight inertial measurement units (IMUs, two for proximal and distal part of each limb), providing both accelerometer and gyroscope assessments, a 3D motion range camera, and an extensible band with programmable resistance, exerted by an electric motor. This arrangement allows to detect and record the movement patterns of the subjects. A Digital Support System (DSS) in cloud allows the remote definition and supervision of exercising programs on a deferred basis, thus reducing operator time. An assisted interaction with subjects is allowed through a chatbot and a smartphone app allowing not only to monitor completed activities and achieved progresses but also to receive reminders and advice for the upcoming planning activities.

Results: The game interface and an avatar of the subject, detected by a range camera, are shown on a TV screen. Operating together, IMUs and the range camera can not only track the movement, but also assess it (as speed and joint range of motion). The DSS operates in an asynchronous mode allowing the continuous remote supervision of the rehabilitation activities, with the possibility to modify the rehabilitation program and progressively adapting it to subject's performance. The chatbot and the smartphone app allow continuous and assiduous first level contact with the subject.

Discussion: The prototype of the platform is innovative as it is based on algorithms both for evaluations and classification of the results data on a deferred basis at variance with the existing telerehabilitation devices. The platform provides also rehabilitation assistance and advice, likely reducing the need of repeated consultations with the physiotherapist. Finally, this intervention technology might take part in the creation of a standardized evaluation and rehabilitation prescription system, leading to better quality of healthcare assistance.

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Characteristics of landings, rotations, and impacts of real-world falls captured by wearable sensors in older persons

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Introduction: Falls are a significant health problem. Approximately 30% of adults 65 and older fall each year. Understanding the characteristics of fall patterns, including landing, impact, and related injuries, is of paramount importance to understand why and when injuries happen and implement adequate interventions to reduce the risk of adverse consequences. We analyzed real-world falls of older persons living in different settings to characterize the landing positions, related impacts, and injuries.

Methods: We analyzed real-world falls from the Farseeing dataset [1], recorded by a wearable sensor worn on the lower back. We selected only those falls with a clear upright pre-fall position. We processed Accelerations and angular velocities to estimate landing positions and the amount of rotation during the descent phase. We computed the prevalence of different types of landings and associated these quantitative characteristics with the reported injuries. Finally, we estimated the impact force using the body mass of the subject and the maximum value of the acceleration norm.

Results: We analyzed 92 falls in total. The landing positions were divided into backwards (BW), forwards (FW), and sideways (SW). The majority of falls were BW (67%), followed by SW (25%) and FW (8%), in accordance with [2]. BW falls were the ones with fewer injuries in percentage, also in accordance with [2]. Only two serious injuries happened, one in FW and one in SW. The highest rotation was found around the vertical direction V (see Fig. 1.a)). Estimated impact forces were between 1000 and 5000 Newton, with a slightly higher impact force for falls with backward landing.

Discussion: Through automatic analysis of wearable sensor data, it was possible to identify different patterns of real-world falls of older people, getting novel insights about the prevalence of different landing positions, associated injuries, movements during descent (i. e., rotations), and fall impact forces. The backward landing was the most frequent landing position, with the lowest relative number of associated injuries (both minor and serious injuries).

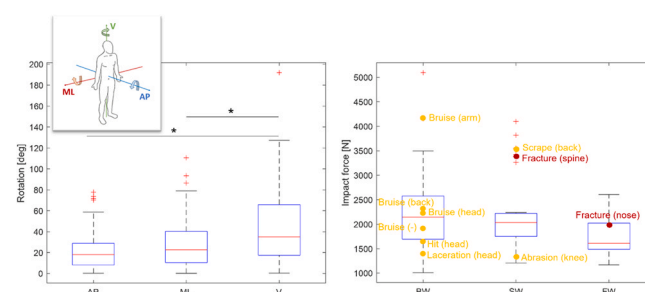


Fig. 1. a) Degrees of rotation during falls around the mediolateral (ML), anteroposterior (AP), and vertical (V) axes, b) Impact forces and injuries.